to mention ensuring operational readiness of nuclear weapons, is all-consuming. The last thing any launch officer needs to worry about is making any autonomous decisions about when to use his nukes. That means having absolute confidence in the communications system that tells him what to do. A former US Navy ballistic missile submarine commander who requested anonymity due to his current job recalled that his number one priority was to stay in constant communication with shore-based nuclear authorities. That is obviously of critical importance in the seaborne submarine fleet, but it is hardly less important on land. Dropped phone calls and network interruptions are an annoyance at normal times, but during a crisis, such failures could inadvertently unleash a nuclear strike. However Kim decides to send the orders for strategic operations, a former senior US nuclear commander notes, they likely would come through communications systems including the country's fiber optics network, as well as occasional line-of-sight radio transmitters to the road-mobile launchers. But how reliable will such systems be?

And all these uncertainties are magnified a dozenfold when talking about sea-based nuclear systems. Perhaps tempting the nuclear gods, Pyongyang apparently desires to develop an indigenous ballistic missile submarine, which is one of the most technologically complex weapons systems in existence.[14] While it remains years away from having such a capability, operating missile submarines would tax North Korea's untested command and control systems in even more acute ways, not least in the absolute confidence in the stability and reliability of the submarine's senior officers.

VI.

Outside of Moscow, in May 2017, Stanislav Petrov died at the age of seventy-seven.[15] Largely unknown to the world, Petrov may have single-handedly averted a nuclear war between the Soviet Union and the United States. As a lieutenant colonel in Soviet air defense headquarters,